GLUING RING MAPS GIVEN AT FINITELY MANY MAXIMAL IDEALS WHOSE KERNELS AND COKERNELS ARE KILLED LOCALLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-chow/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The two cross-chapter references (a lemma and a remark of More on Algebra) are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Formal glueing of modules and algebras over a finite product of localizations. More on Algebra, Lemma more-algebra-lemma-application-formal-glueing (canonical Stacks tag 0ALK) and More on Algebra, Remark more-algebra-remark-formal-glueing-algebras (canonical Stacks tag 05EU); invoked at proof lines 548-555. Supply the existence of the A-module map and the algebra structure on the glued module; the reduction of the hypothesis to their input, i.e. the flatness of A -> A' with A/I = A'/IA', is inside the counted span.

Source reference. Original lemma 0ALK: sources/stacks-chow/tags.txt, source lines 13373–13373. Cross-chapter standard prerequisite invoked once at proof lines 548-550 to glue the underlying module map; the verification that its hypotheses hold for A -> prod A_{m_i} is retained in the counted span.

Source reference. Original remark 05EU: sources/stacks-chow/tags.txt, source lines 7003–7003. Cross-chapter standard prerequisite invoked once at proof lines 552-555 to endow the glued module with an A-algebra structure; it supplies the algebra structure only, while the reading off of finiteness, injectivity and surjectivity is retained in the counted span.

Lemma 0.1. *Let A be a Noetherian ring. Let $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ be pairwise distinct maximal ideals of A. For $i = 1, \ldots, r$ let $\varphi_i : A_{\mathfrak{m}_i} \to B_i$ be a ring map whose kernel and cokernel are annihilated by a power of $\mathfrak{m}_i$. Then there exists a ring map $\varphi : A \to B$ such that*

(1) *the localization of φ at $\mathfrak{m}_i$ is isomorphic to φ_i, and*
(2) $\mathrm{Ker}(\varphi)$ *and* $\mathrm{Coker}(\varphi)$ *are annihilated by a power of* $\mathfrak{m}_1 \cap \ldots \cap \mathfrak{m}_r$.

Moreover, if each φ_i is finite, injective, or surjective then so is φ.

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Proof. Set $I = \mathfrak{m}_1 \cap \ldots \cap \mathfrak{m}_r$. Set $A_i = A_{\mathfrak{m}_i}$ and $A' = \prod A_i$. Then $IA' = \prod \mathfrak{m}_i A_i$ and $A \to A'$ is a flat ring map such that $A/I \cong A'/IA'$. Thus we may use More on Algebra, Lemma 0ALK to see that there exists an A-module map $\varphi : A \to B$ with φ_i isomorphic to the localization of φ at $\mathfrak{m}_i$. Then we can use the discussion in More on Algebra, Remark 05EU to endow B with an A-algebra structure matching the given A-algebra structure on B_i. The final statement of the lemma follows easily from the fact that $\mathrm{Ker}(\varphi)_{\mathfrak{m}_i} \cong \mathrm{Ker}(\varphi_i)$ and $\mathrm{Coker}(\varphi)_{\mathfrak{m}_i} \cong \mathrm{Coker}(\varphi_i)$. $\square$